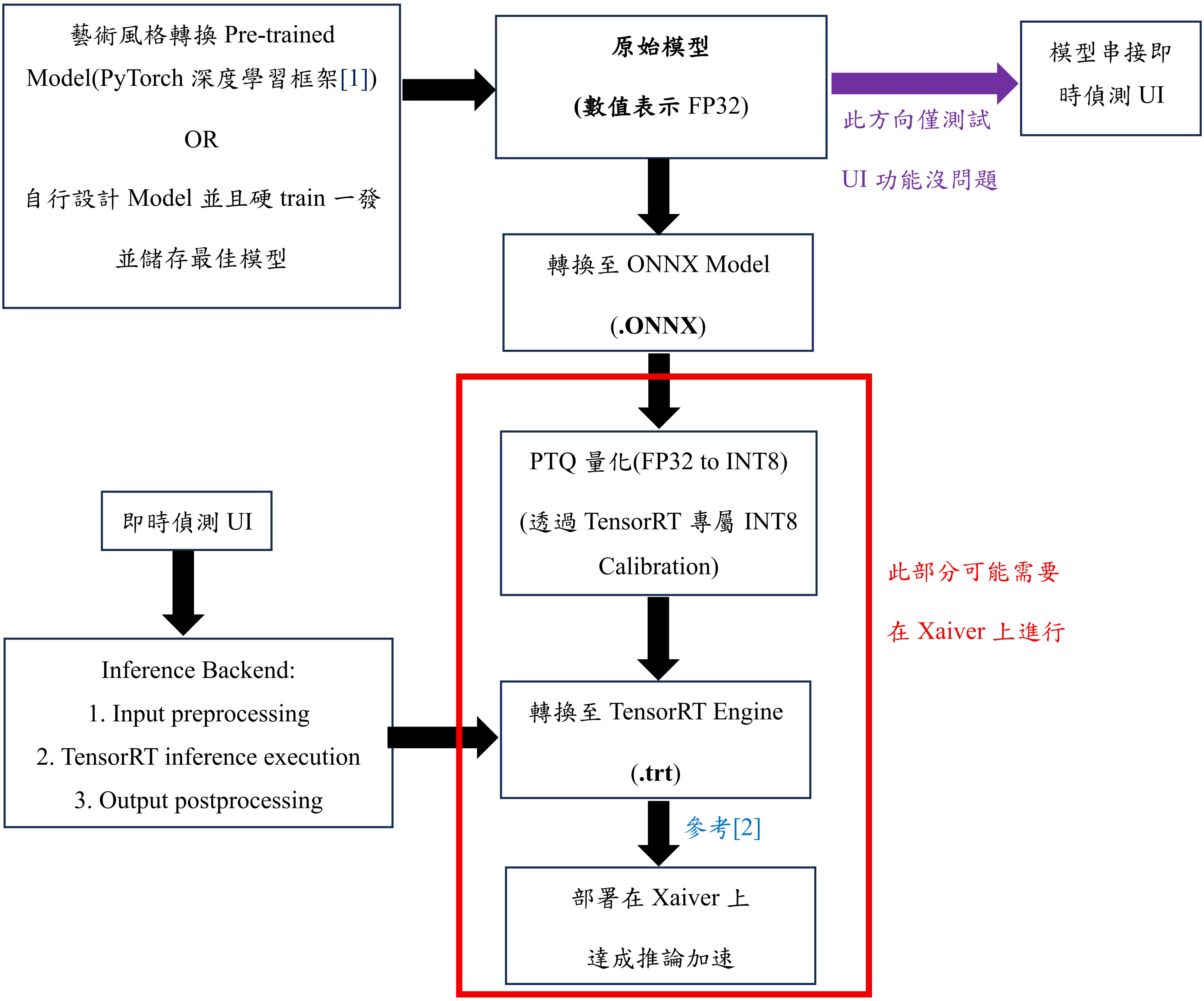
Project 2: A Real-Time Style Transfer System

Description: 即時影像結合藝術風格轉換並部屬在 Xaiver 邊緣裝置上

Pipeline Design:



[1] DL 模型權重檔案格式:

https://547labrecord.blogspot.com/2025/02/blog-post.html

[2] PyTorch model to TensorRT for 3-8x speedup:

https://hackmd.io/_oaJhYNqTvyL_h01X1Fdmw?view

Model Compression: Quantization

考量裝置在運算資源有限且暫不追求精度情況下，推薦使用 **Post-Training Quantization** ，在數值表示方面，一般深度學習模型是以 FP32 表示，考慮最終部署在邊緣裝置以利於後續推論加速和部屬，需映射至 INT8。

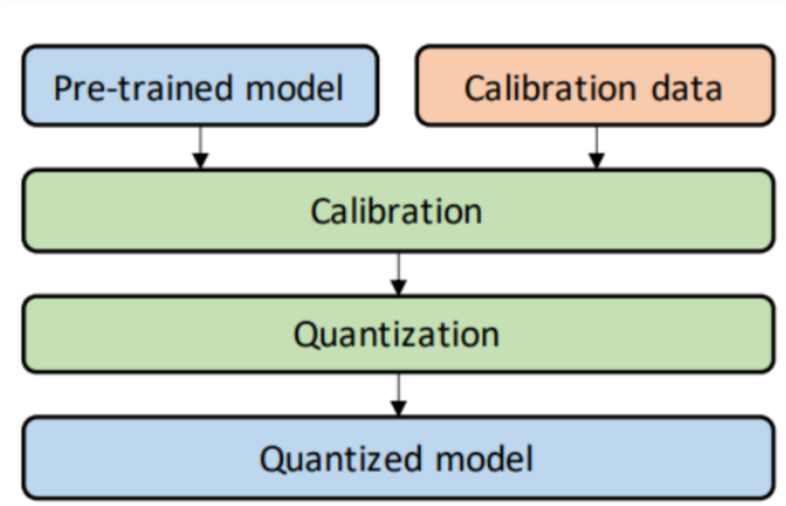
Jetson Xavier 部屬使用量化相關資料(高度正相關):

https://hackmd.io/@YungHuiHsu/Sy8ZfM1r3

兩大量化方法 Pipeline:

6.1 Post-Training Quantization (PTQ)

In PTQ, quantization is applied **after** the model has already been trained in floating point.



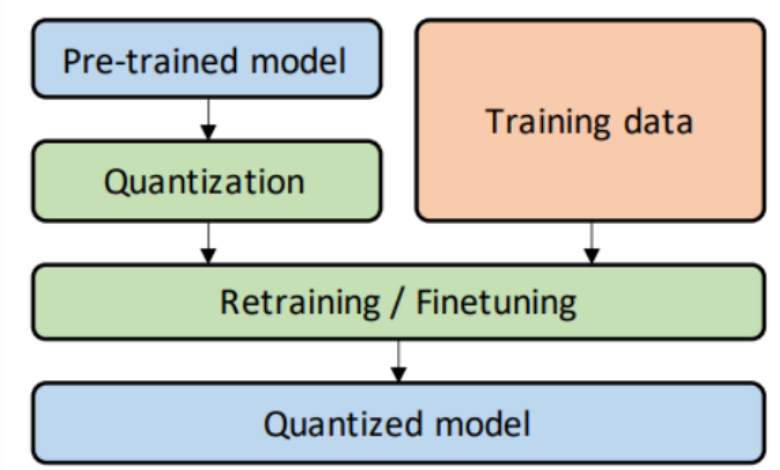
The trained model's weights and activations are directly mapped to lower precision (e.g., INT8) without re-training. Calibration data is often used to determine optimal scaling factors.

PTQ is suitable when **training data or computational resources are limited**, but may not meet accuracy requirements in all scenarios.

> ⓘ Note
>
> - Pros: Fast and easy to apply which no need for additional training. Besides, PTQ is well supported by most ML frameworks (TensorFlow Lite, PyTorch, etc.).
> - Cons: Accuracy degradation may be significant, especially for complex models like transformers or models with sensitive layers.

6.2 Quantization-Aware Training (QAT)

QAT addresses the accuracy drop problem of PTQ by incorporating quantization **during training**.



During forward propagation, fake quantization is inserted: weights and activations are quantized and then dequantized, so the network "sees" the effect of quantization. During backward propagation, gradients are computed with special techniques to allow training despite quantization being non-differentiable.

QAT is the preferred choice when **high accuracy must be maintained on constrained hardware**.

> ⓘ Note
>
> - Pros: Provides much **higher accuracy** compared to PTQ, often close to full-precision models. Enables deployment on integer-only hardware without retraining.
> - Cons: Requires access to the training data and **more computational cost**. Training is more complex and time-consuming.

Source:

https://hackmd.io/@2025eai/BkGdON-3gx/https%3A%2F%2Fhackmd.io%2F%402025eai%2Fr1ZLeJv9ll

Hardware Specification

	Jetson Xavier NX series	
	Jetson Xavier NX (16GB)	Jetson Xavier NX (8GB)
AI Performance	21 TOPS	
GPU	384-core NVIDIA Volta™ architecture GPU with 48 Tensor Cores	
GPU Max Frequency	1100 MHz	
CPU	6-core NVIDIA Carmel Arm®v8.2 64-bit CPU 6MB L2 + 4MB L3	
CPU Max Frequency	1.9 GHz	

Source:

https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-xavier-series/

Jetson Xavier NX Developer Kit User Guide:
https://developer.nvidia.com/embedded/secure/tools/files/jetpack-sdks/jetpack-4.4-ga/jetson_xavier_nx_developer_kit_user_guide.pdf

Xaiver pip-list:

https://drive.google.com/file/d/1uw67sh4OK1u-teaiVK476A-jllfj-kFh/view?usp=sharing

虛擬環境安裝和套件相容(本地端訓練並考量最終部屬環境)

說明: 考慮 Win11 及 Xaiver 安裝套件環境相容性問題

Source: https://gemini.google.com/share/ac4311f99abc

Xaiver 環境資訊

查詢指令: jtop

Platform:

Machine: aarch64

System: Linux

Distribution: Ubuntu 20.04 focal

Release: 5.10.104-tegra

Python: 3.8.10

Libraries:

CUDA: 11.4.315

cuDNN: 8.6.0.166

TensorRT: 8.5.2.2

VPI: 2.2.7

Vulkan: 1.3.204

OpenCV: MISSING

Hardware:

Model: NVIDIA Jetson Xavier NX Developer Kit

Module: NVIDIA Jetson Xavier NX (Developer kit)

L4T: 35.3.1

Jetpack: 5.1.1